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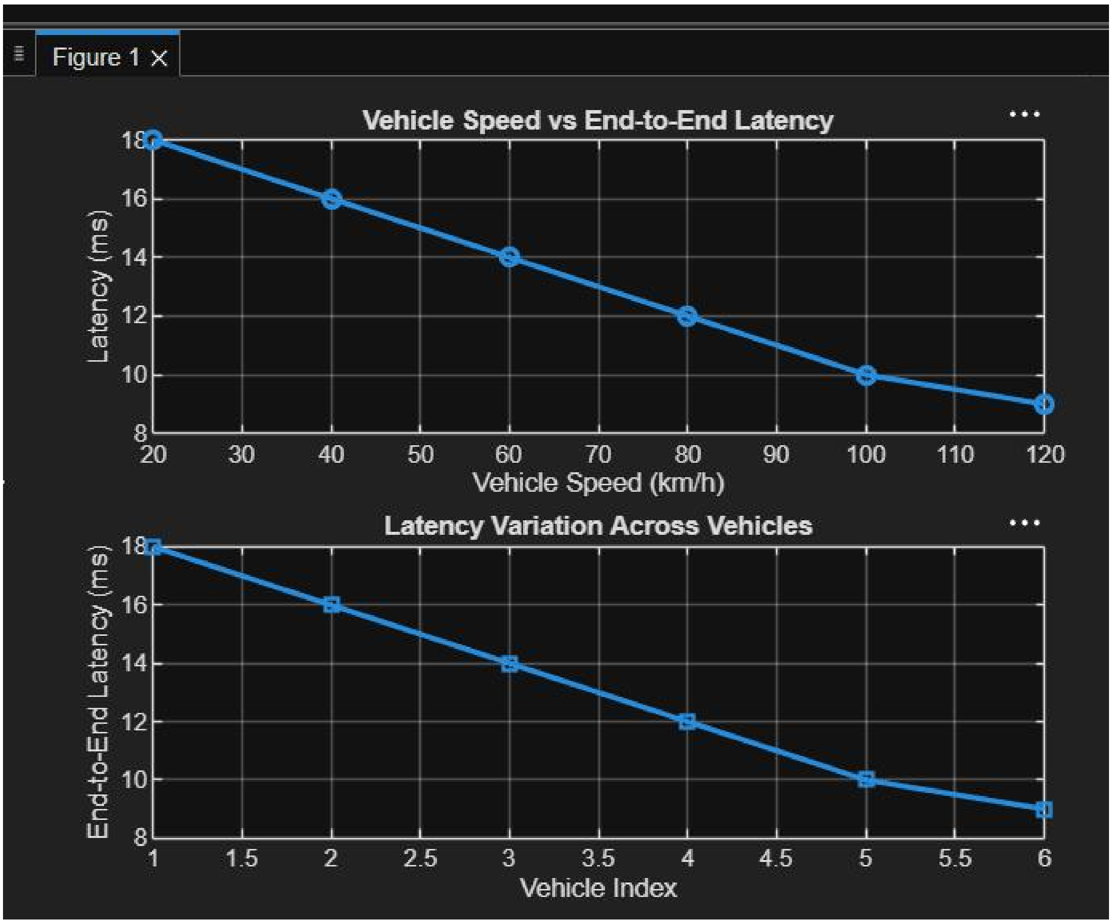
Ex.No. 13: V2X Latency Comparison

OBJECTIVE 1:

CODE:

```
clc;
clear;
close all;
% Vehicle Speed (km/h)
speed = 20:20:120;
% Simulated End-to-End Latency (ms)
latency = [18 16 14 12 10 9];
disp('Vehicle Speed (km/h) End-to-End Latency (ms)')
disp([speed' latency'])
figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(speed,latency,'-o','LineWidth',2)
title('Vehicle Speed vs End-to-End Latency')
xlabel('Vehicle Speed (km/h)')
ylabel('Latency (ms)')
grid on
% -------- Graph 2 --------
```

```
subplot(2,1,2)
plot(1:length(latency),latency,'-s','LineWidth',2)
title('Latency Variation Across Vehicles')
xlabel('Vehicle Index')
ylabel('End-to-End Latency (ms)')
grid on
```



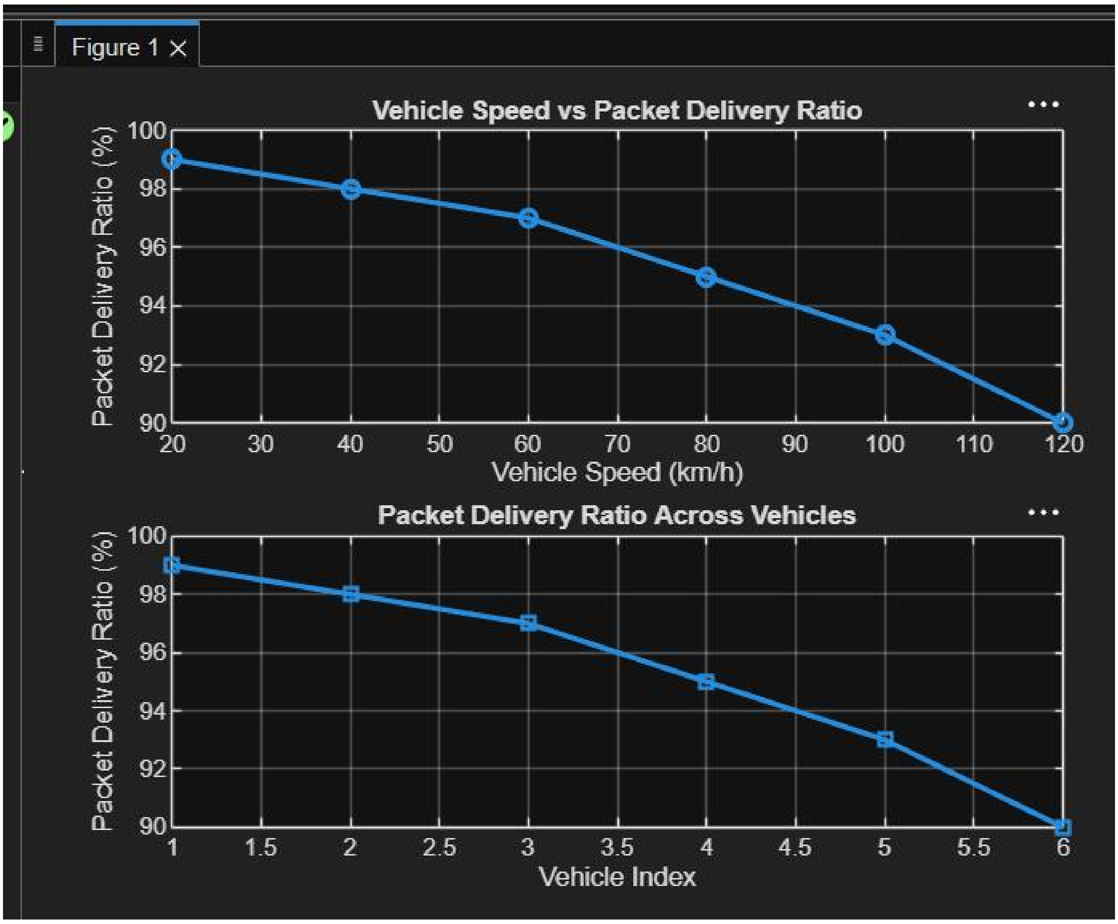
OBJECTIVE2:

CODE:

```
clc;
clear;
close all;
```

```
% Vehicle Speed (km/h)
speed = 20:20:120;
% Simulated Packet Delivery Ratio (%)
PDR = [99 98 97 95 93 90];
disp('Vehicle Speed (km/h) Packet Delivery Ratio (%)')
disp([speed' PDR'])
figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(speed,PDR,'-o','LineWidth',2)
title('Vehicle Speed vs Packet Delivery Ratio')
xlabel('Vehicle Speed (km/h)')
ylabel('Packet Delivery Ratio (%)')
grid on

% -------- Graph 2 --------
subplot(2,1,2)
plot(1:length(PDR),PDR,'-s','LineWidth',2)
title('Packet Delivery Ratio Across Vehicles')
xlabel('Vehicle Index')
ylabel('Packet Delivery Ratio (%)')
grid on
```



OBJECTIVE3:

CODE:

```
clc;
clear;
close all;

% Vehicle IDs
vehicle = 1:6;

% Vehicle Speed (km/h)
speed = [30 45 60 75 90 105];
```

```
% Communication Range (m)
range = [280 260 240 220 200 180];

% Display values
disp('Vehicle  Speed(km/h)  Communication Range(m)')
disp([vehicle' speed' range'])

figure

% -------- Graph 1 --------
subplot(2,1,1)
plot(vehicle, speed, '-o', 'LineWidth', 2)
title('Vehicle Speed of Different Vehicles')
xlabel('Vehicle Index')
ylabel('Vehicle Speed (km/h)')
grid on

% -------- Graph 2 --------
subplot(2,1,2)
plot(speed, range, '-s', 'LineWidth', 2)
title('Vehicle Speed vs Communication Range')
xlabel('Vehicle Speed (km/h)')
ylabel('Communication Range (m)')
grid on
```

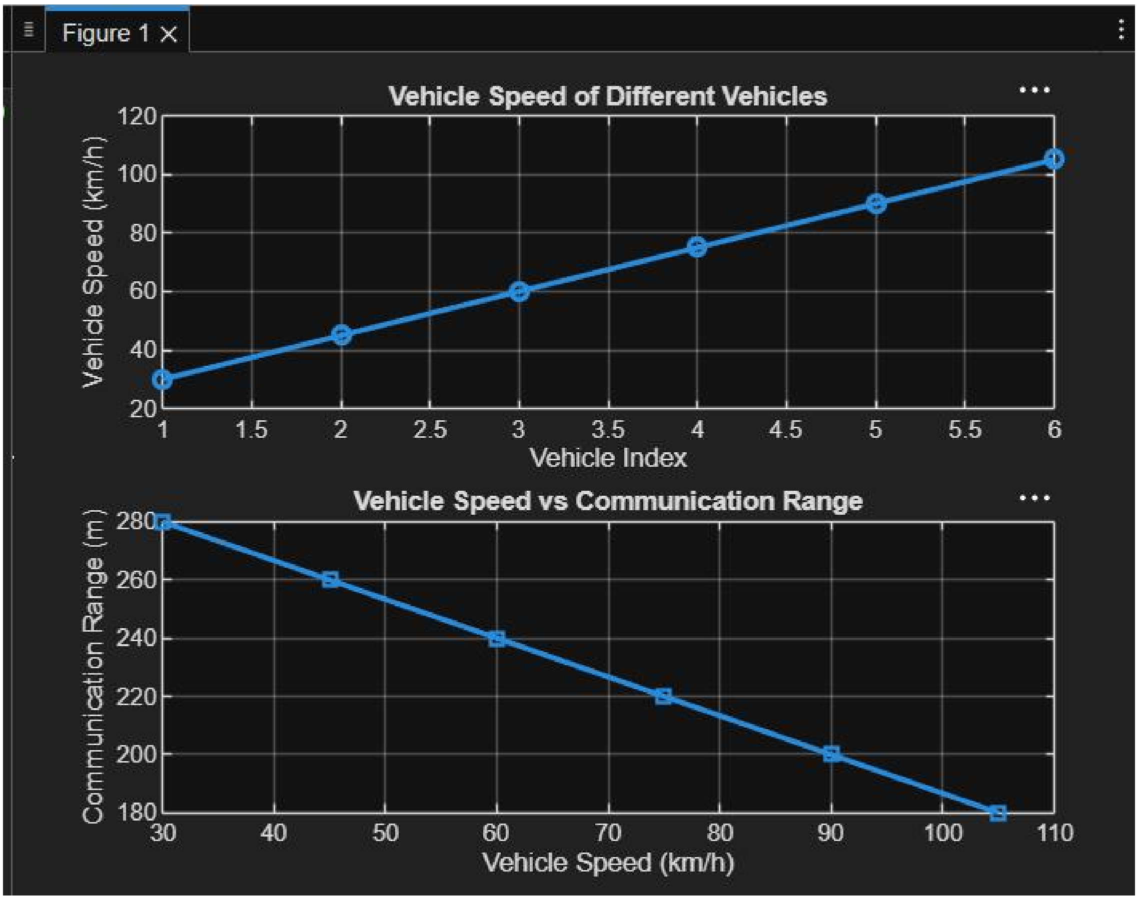


Figure 1
Vehicle Speed of Different Vehicles
Vehicle Speed (km/h)
120
100
80
60
40
20
1
1.5
2
2.5
3
3.5
4
4.5
5
5.5
6
Vehicle Index
Vehicle Speed vs Communication Range
Communication Range (m)
280
260
240
220
200
180
30
40
50
60
70
80
90
100
110
Vehicle Speed (km/h)